

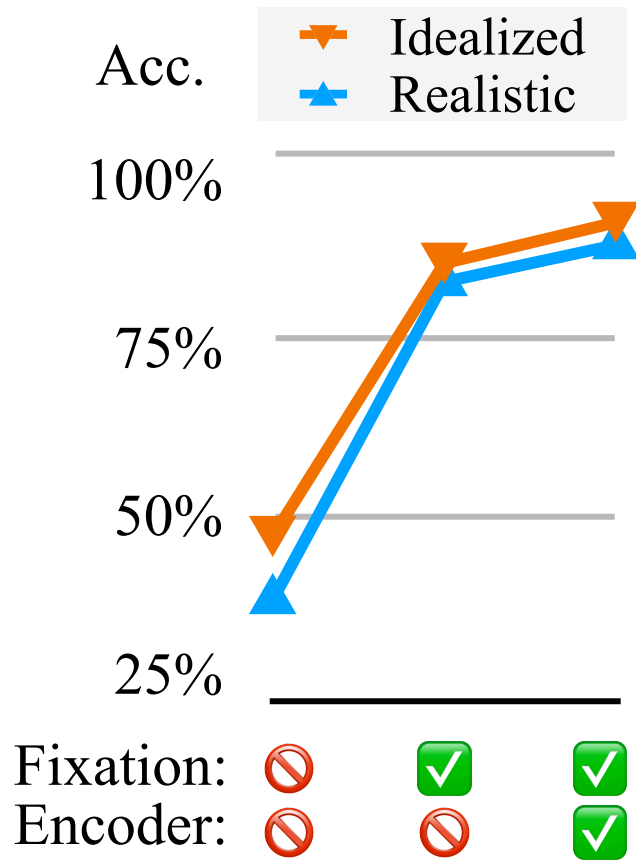
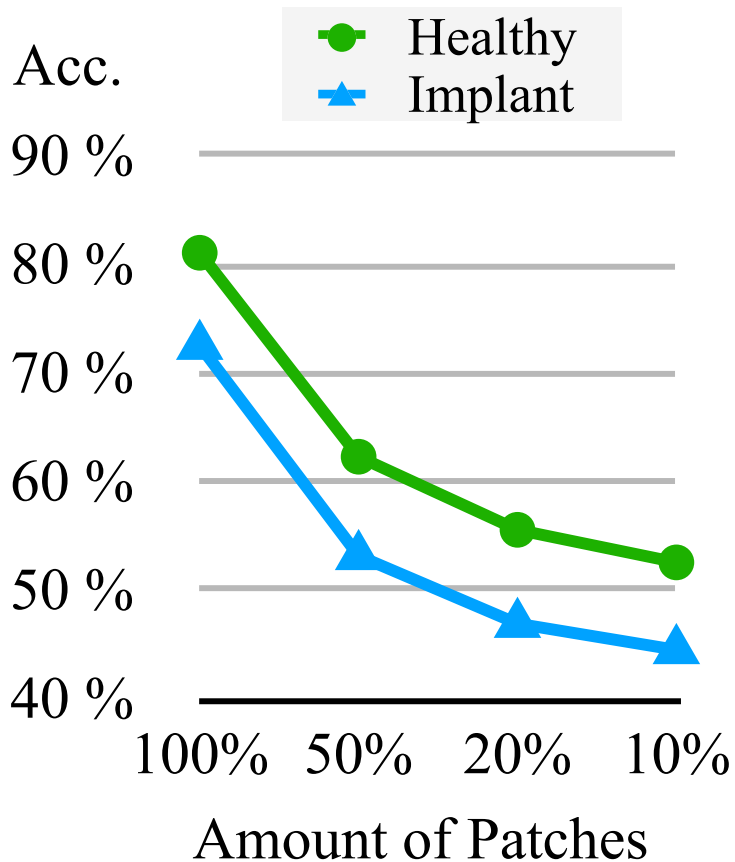
Yuli Multi-Enhancing Artificial Vision Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY

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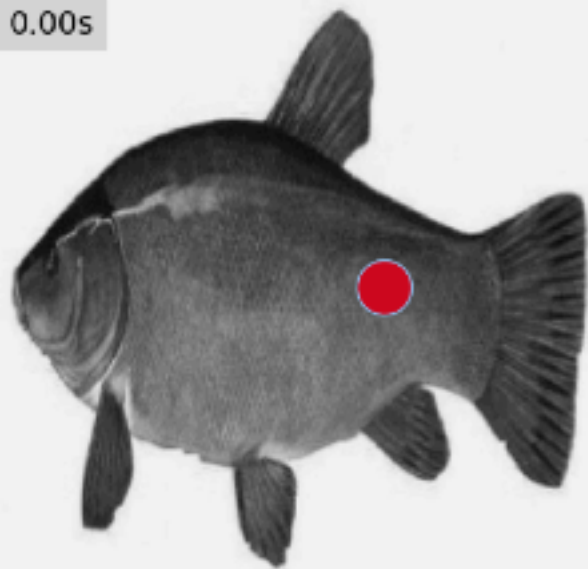
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Visual Fixation-based Simulation



Fixation	Encoder	Linear probing	Accuracy (in %)	
			Param. A	Param. B
$\times$	$\times$	Learnable	47.46	38.70
$\times$	U-Net	Learnable	51.49	40.59
$\checkmark$	$\times$	Learnable	85.20	81.99
$\checkmark$	Linear	Learnable	21.12	21.43
$\checkmark$	U-Net*	Learnable	53.81	68.79
$\checkmark$	U-Net	Frozen	85.94	82.62
$\checkmark$	U-Net	Learnable	90.85	87.72
Healthy upper bound as compare:			92.76	

0.00s



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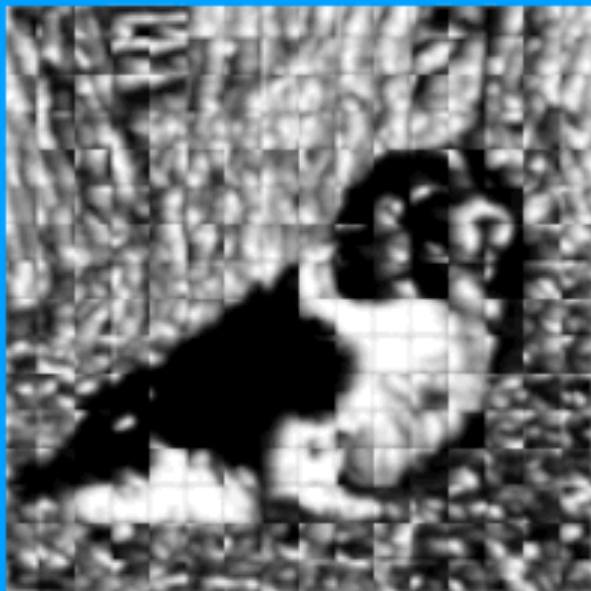


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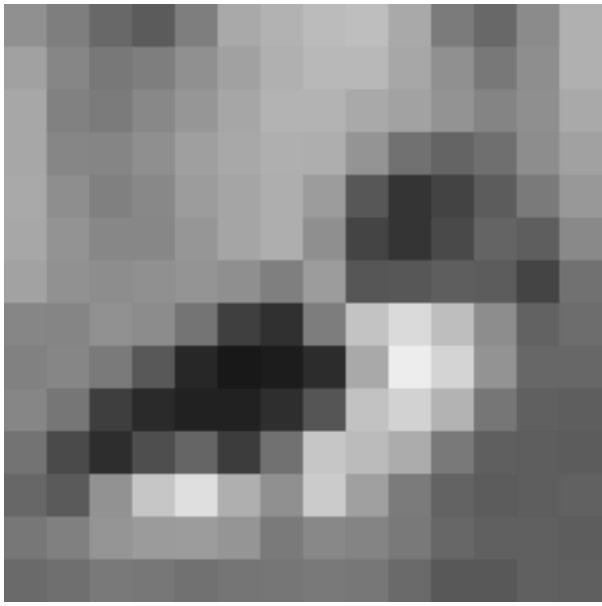


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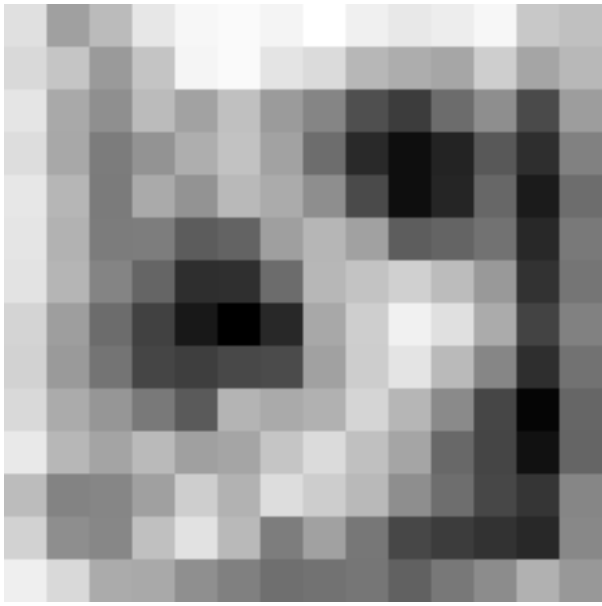








The first part of the paper discusses the importance of maintaining accurate records of all transactions, including sales, purchases, and expenses. This is essential for ensuring the integrity of the financial statements and for providing a clear audit trail. The second part of the paper focuses on the importance of maintaining accurate records of all assets and liabilities, including property, equipment, and accounts payable. This is essential for ensuring the accuracy of the balance sheet and for providing a clear audit trail. The third part of the paper discusses the importance of maintaining accurate records of all income and expenses, including salaries, wages, and interest. This is essential for ensuring the accuracy of the income statement and for providing a clear audit trail. The fourth part of the paper focuses on the importance of maintaining accurate records of all cash flows, including receipts and payments. This is essential for ensuring the accuracy of the cash flow statement and for providing a clear audit trail. The fifth part of the paper discusses the importance of maintaining accurate records of all taxes, including income tax, sales tax, and property tax. This is essential for ensuring the accuracy of the tax returns and for providing a clear audit trail. The sixth part of the paper focuses on the importance of maintaining accurate records of all other financial information, including bank statements, credit card statements, and insurance policies. This is essential for ensuring the accuracy of the financial statements and for providing a clear audit trail.



Downsized

Optimized Stimulus

Optimized Percept

Fixations

Saiiancy

Optimized Percept

[Wuetai., 2025a]

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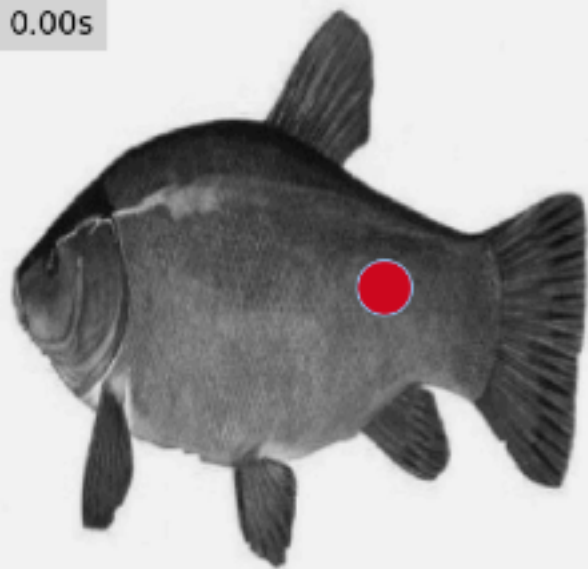
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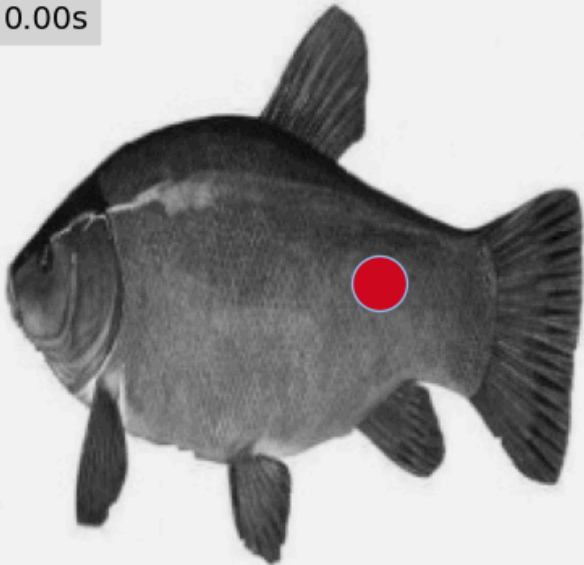
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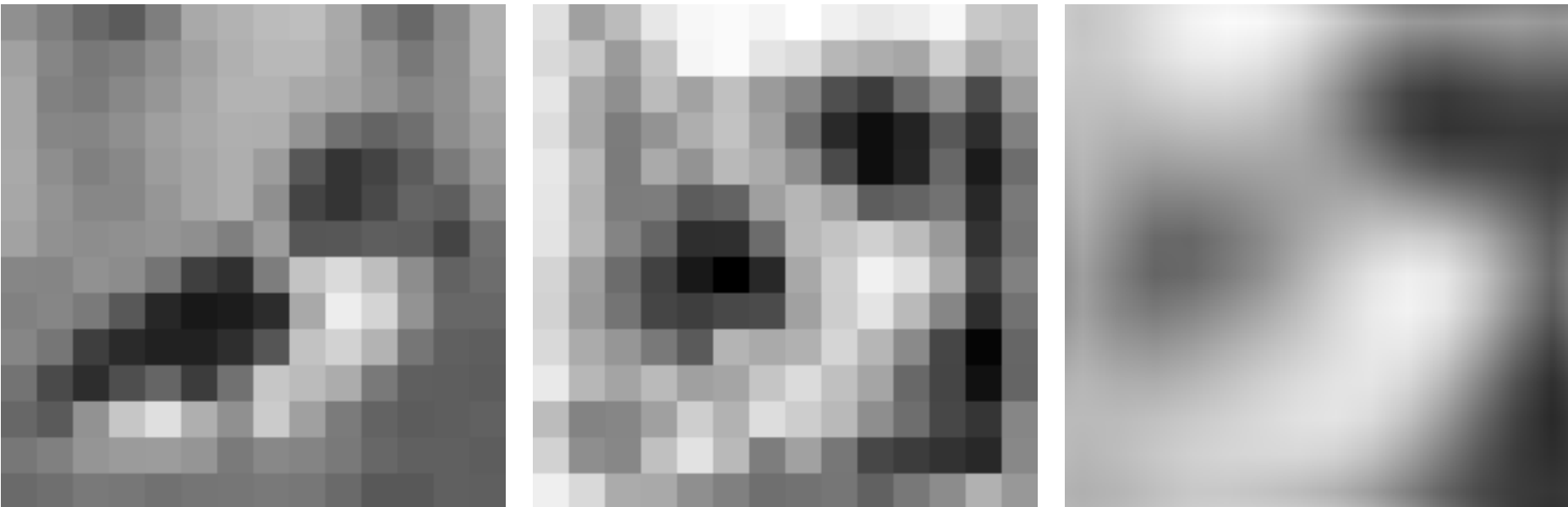






Visual Fixation-based Simulation

Fixation	Encoder	Linear probing	Accuracy (in %)	
			Param. A	Param. B
✗	✗	Learnable	47.46	38.70
✗	U-Net	Learnable	51.49	40.59
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Healthy upper bound as comparate:			92.76	



Downsized

Optimized Stimulus

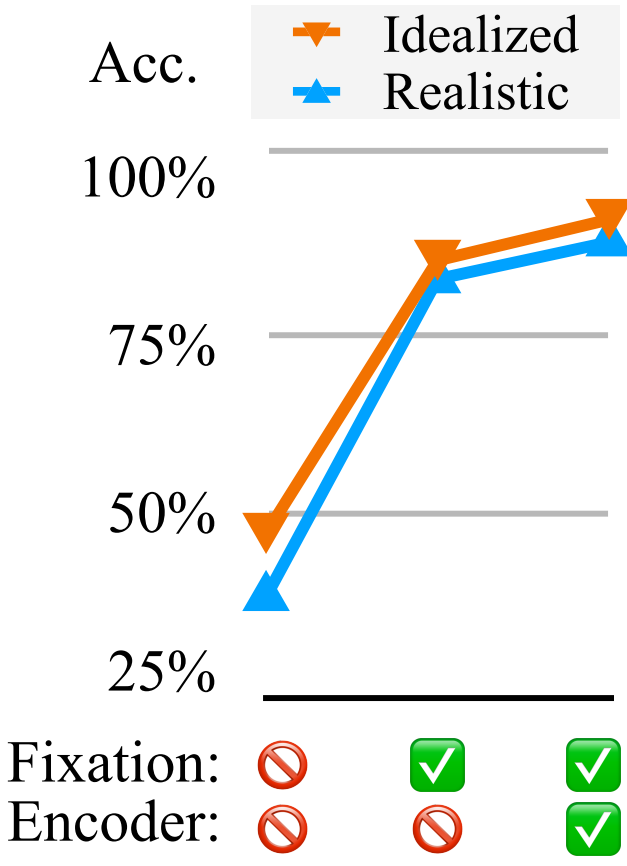
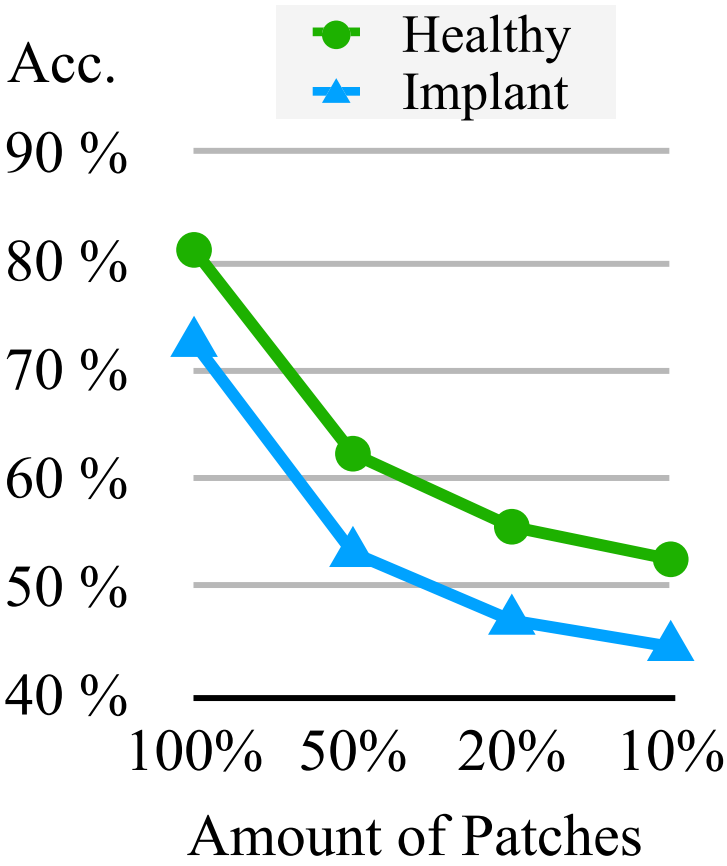
Optimized Percept



Fixations

Saliency

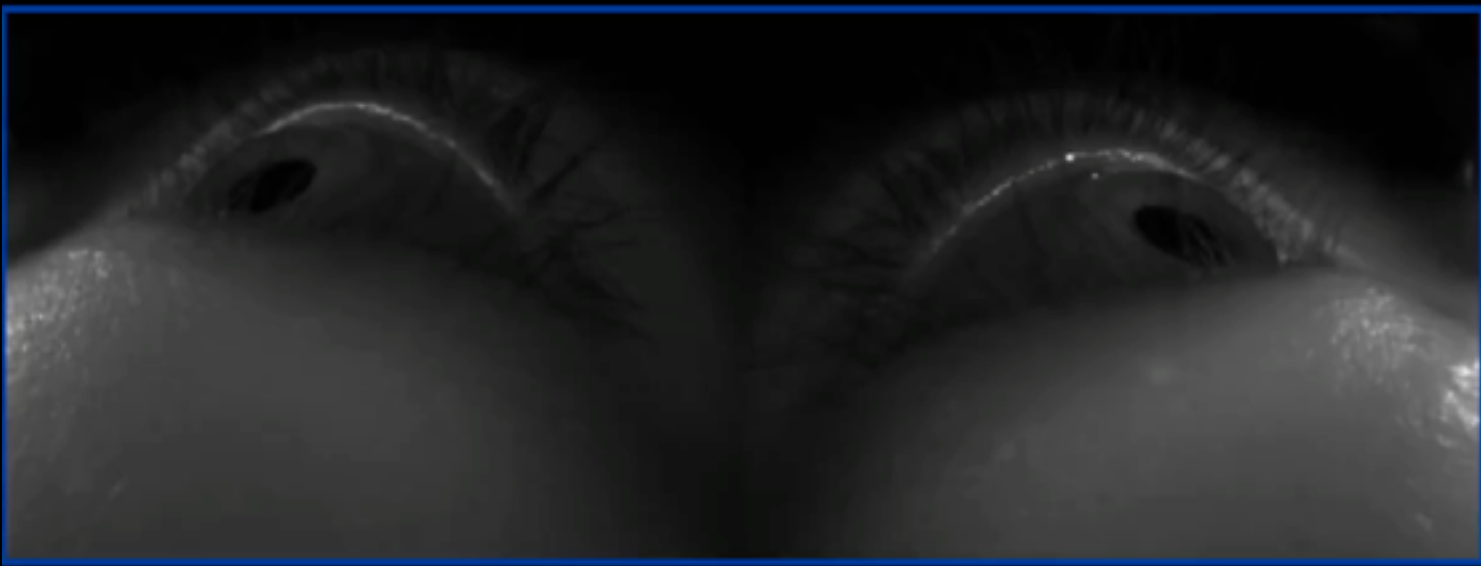
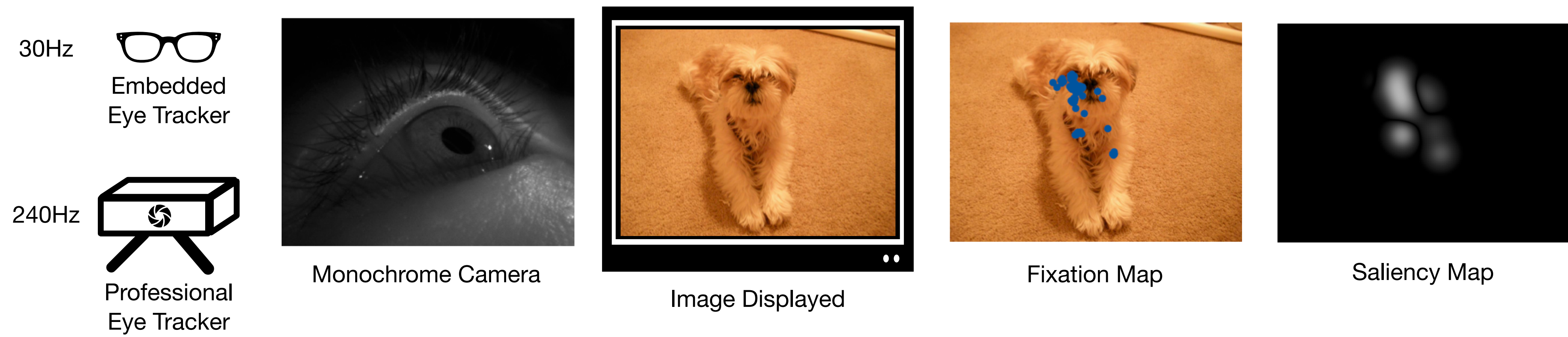
Optimized Percept



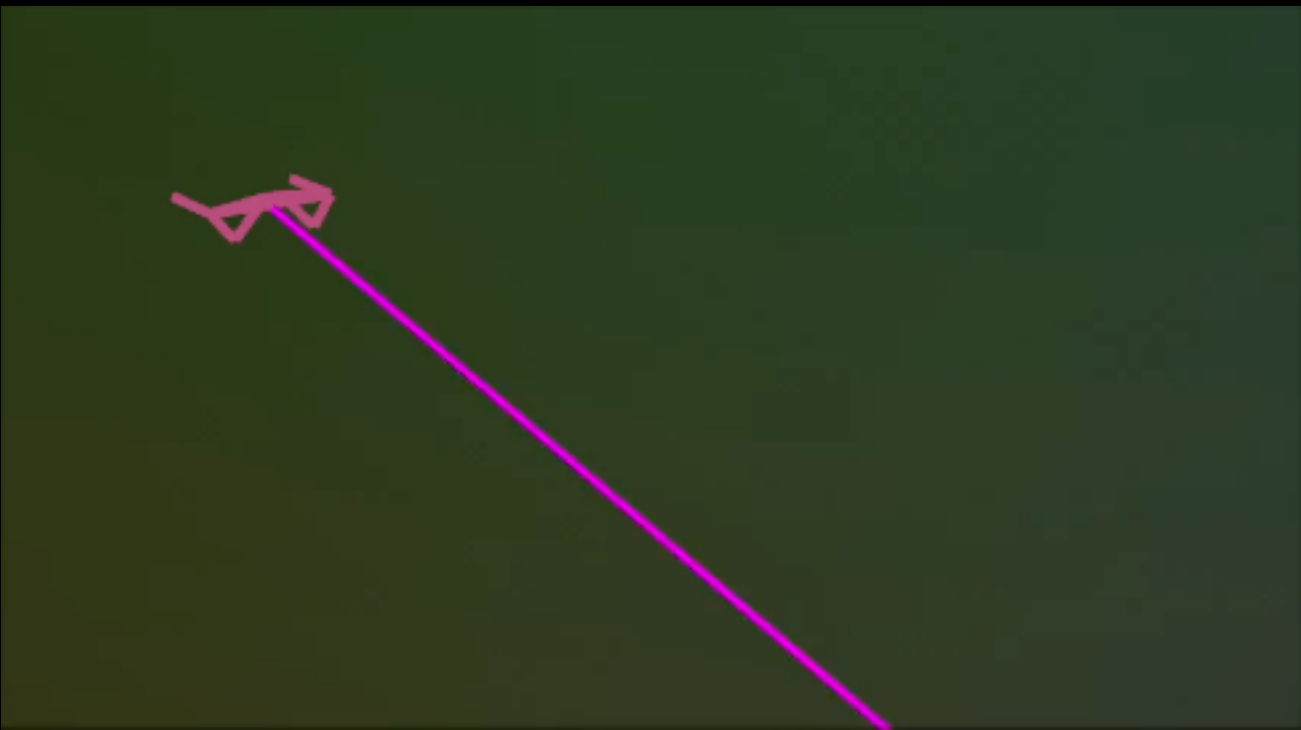
[Wu et al., 2025a]

Visual Fixation Consistency

Ethical approval was granted by the Interfaculty Ethics Committee of RWTH Aachen University with the reference number 03/25.



Eye movements



Gaze direction



Visual fixation

[Wu et al., 2025b]